

GOOD MORNING EVERYONE

We will start @ 10:10 am

3
1
2
3

pairs $\rightarrow a, b$
 fix 1
 $1+1=2 \times$
 $1+2=3 \checkmark$
 $1+3=4 \times$

Fix 2
 $2+1=3 \checkmark$
 $2+2=4 \times$
 $2+3=5 \checkmark$

Fix 3
 $3+1=4 \times$
 $3+2=5 \checkmark$
 $3+3=6 \times$

1 2
2 1
2 3
3 2

0 1 2
1 2 3

$i=0$ $j=0$

1 1
1 2
1 3

for (int i=0; i<n; i++) {
 for (int j=0; j<n; j++) {
 int sum = arr[i] + arr[j];
 if (sum % 2 != 0)
 print (arr[i] + " + " + arr[j]);
 }
}

$\rightarrow$ fix de.

$i=1$ $j=1$
 2 2
 2 3
 2 1

2 1

1 2
1 3

2 3

i, j
 j, i

1 2 3 4

4
1 2
1 4
2 1
2 3
3 2
3 4
4 1
4 3

1 2
2 1
1 4
1

for (int i=0; i<n; i++)
 for (int j=i; j<n; j++)

2D (Two Dimensional) Array:

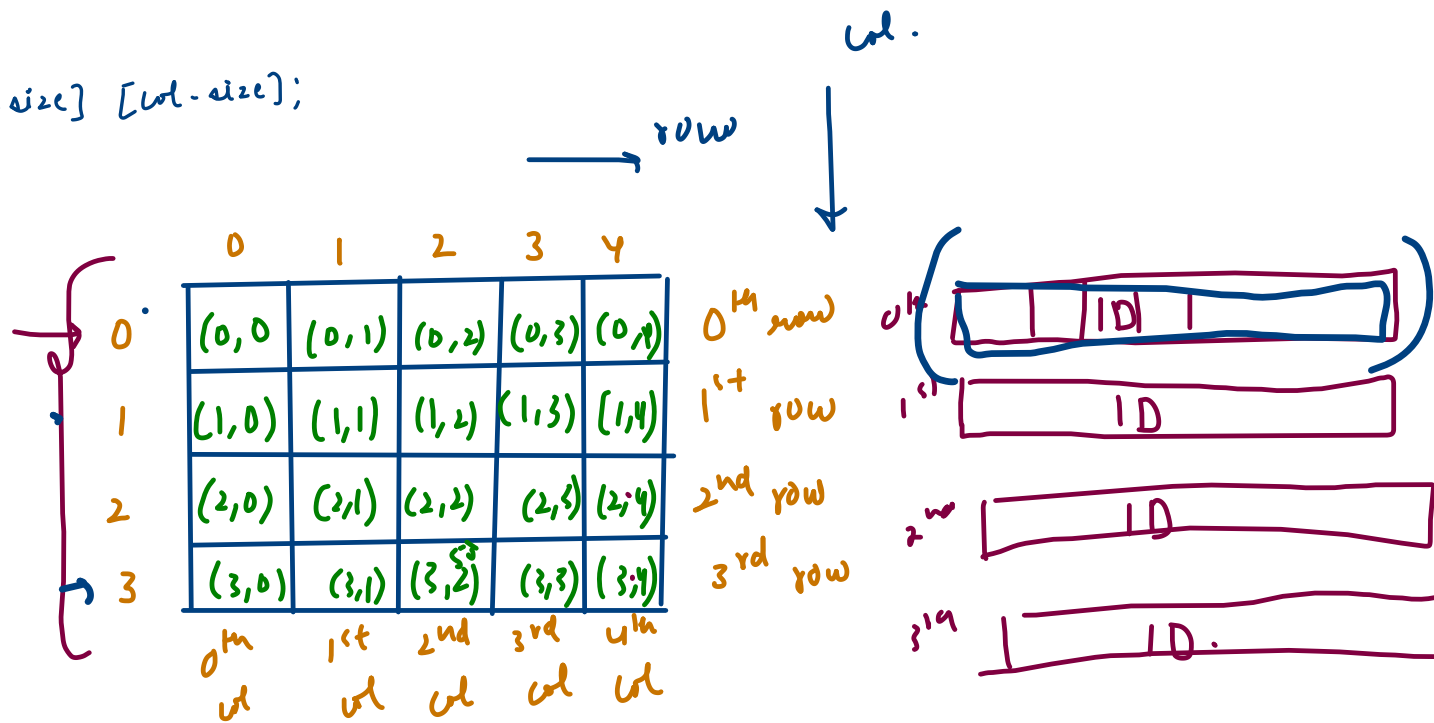
$\uparrow$
int arr[] = new int [n];

Syntax
datatype array-name $\uparrow$ [] [] = new int [row-size] [col-size];

Ex:- int arr [] [] = new int [4] [5]; 4, 5

Total size = row $\times$ col $\Rightarrow 4 \times 5 = 20$
(rowth no.; col no.)

arr[3][2] = 50



$\Rightarrow$ How many rows are there $\Rightarrow$ "arr.length"

$\Rightarrow$ How many cols are there $\Rightarrow$ arr[0].length;

Go to 0th row and get its length

$\downarrow$
no. of cols.

```

public static void main(String[] args) {
    /* Enter your code here. Read input from STDIN. Print output to STDOUT */
    Scanner scn = new Scanner(System.in);
    int n = scn.nextInt(); // Number of rows 3
    int m = scn.nextInt(); // Number of cols 3

    for(int i=0; i<n; i++) {
        for(int j=0; j<m; j++) {
            arr[i][j] = scn.nextInt();
        }
    }
}

```

	0	1	2
0	7	1	2
1	9	8	6
2	3	4	5

$i=0 < 3$ (T)

$j=0 < 3$ (T) $arr[0][0] = 7$

$j=1 < 3$ (T) $arr[0][1] = 1$

$j=2 < 3$ (T) $arr[0][2] = 2$

$j=3 < 3$ (F)

$i=1 < 3$ (T)

$j=0 < 3$ (T) $arr[1][0] = 9$

$j=1 < 3$ (T) $arr[1][1] = 8$

$j=2 < 3$ (T) $arr[1][2] = 6$

$j=3 < 3$ (F)

$i=2 < 3$ (T)

$\hookrightarrow j=0 < 3$ (T) $arr[2][0] = 3$

$\hookrightarrow j=1 < 3$ (T) $arr[2][1] = 4$

$\rightarrow j=2 < 3$ (T) $arr[2][2] = 5$

$\hookrightarrow j=3 < 3$ (F)

$i=3 < 3$ (F)